

Patrick Smyth CV

EDUCATION

Ph.D. in English, Graduate Center of the City University of New York (CUNY), 2021
Faculty Award for Most Distinguished Dissertation of 2021

Double B.A. in English and German Literature, Kenyon College, Ohio
magna cum laude, 2009
Highest Honors in English (Honors thesis)

Certified Instructor, [Software Carpentry](#), 2017

Certificate, [Interactive Technology and Pedagogy \(ITP\)](#), 2016

POSITIONS

Principal Developer Relations Engineer
[Chainguard](#)
October 2025 — present

- Influence revenue through vehicles such as Learning Lab, the Assemble 2025 customer conference, and 1-1 work with customers; have driven \$16.5m ARR to date.
- Present at conferences including PyCon, RSAC, and SwampUP. All accepted through organic community CFP processes, not sponsored.
- Work flexibly to fill perceived gaps, including pinch-hitting as a PM on AI initiatives, jumping on customer calls to provide domain expertise, and leading technical development for a critical website.
- Develop extensive content in the form of videos, LinkedIn posts, course modules, deep dives, tutorials, and breakout sessions.

Staff Developer Relations Engineer
[Chainguard](#)
March 2024 — October 2025

Chief Learner / Principal Consultant
[Iota School](#)
August 2021 — March 2025

- Received a development grant from Pandas / NumFOCUS to document, lead, and record a five-workshop series on data science fundamentals and sonification with over

400 participants.

- Worked with a Jupyter core developer and research astronomers to improve web-based notebook exports at James Webb and MAST; presented on the project at JupyterCon 2023 in Paris.
- Replatformed the docs site and updated documentation for Sigstore as a Google Season of Docs technical writer.
- Created a series of tutorials on deploying neural networks for object recognition as demo materials for a data science platform under development at Enthought.

Postdoctoral Fellow in Humanities Entrepreneurship

[Publics Lab](#)

The Graduate Center, CUNY

August 2021 — August 2022

Research Software Engineer, [Foundations for Research Computing](#)

Columbia University

January 2019 — August 2021

- Managed a team of staff, faculty, and graduate student instructors to teach over 85 technical workshops for 3,000+ researchers.
- Developed new programming including a Curriculum Innovation Grant, intermediate intensives on NLP and ML in cooperation with Google Research, and a popular Python User Group.
- Worked with Columbia's Data Science Institute, X (formerly Google X), and the TensorFlow team to develop technical training.
- Led workshops on UNIX, scraping, data science fundamentals, visualization, accessing APIs, and ML with scikit-learn.

Advisor, [MA Program in Digital Humanities](#)

The Graduate Center, CUNY

September 2017 — December 2018

Lead Curriculum Designer, Digital Humanities Research Institute

[Institutes for Advanced Topics in the Digital Humanities](#), NEH grant

September 2017 — August 2018

[View Website](#)

[View Curriculum](#)

[#DHRI on Twitter](#)

Digital Initiatives Fellowship, GC Digital Scholarship Lab
CUNY Graduate Center
September 2014 — August 2017

Developer and Documentation Specialist, [DH Box](#) NEH Startup Grant
The Graduate Center, CUNY
September 2014 — August 2018

WRITING

"Mythos Pulls Zero-Days Forward: Here's What You Need to Know Now"
Chainguard Unchained, April 2026
[Read article](#)

"2026: The Year of AI-Assisted Attacks"
Chainguard Unchained, April 2026
[Read article](#)

"Attacks Rewritten: Where Malware Enters the Build" (with Manfred Moser)
Chainguard Unchained, April 2026
[Read article](#)

"Going Deep: Upstream Distros and Hidden CVEs"
Chainguard Unchained / Chainguard Research, February 2026
[Read article](#)

"The Maturity Gap in ML Pipeline Infrastructure"
Chainguard Unchained, January 2026
[Read article](#)

"The Only Rule: Don't Look at the Code" (Chainguard Vibelympics retrospective)
Chainguard Unchained, December 2025
[Read article](#)

"Deep Dive: Where Does Grype Data Come From?"
Chainguard / dev.to, November 2024
[Read article](#)

"Securely Containerize a Python Application with Chainguard Images"
Chainguard / dev.to, April 2024
[Read article](#)

"Double Take: A Memoir." Couser, G. Thomas, and Susannah B. Mintz, editors. Disability Experiences: Memoirs, Autobiographies, and Other Personal Narratives. 2019. (review)

"Creating Web APIs with Python and Flask"

Programming Historian, 2018

[Read article](#)

"DH Box and Access in the Digital Humanities"

dh+lib, 2017

[Read article](#)

"Ebooks and the Digital Paratext: Emerging Trends in the Interpretation of Digital Media"
in Examining Paratextual Theory and its Applications in Digital Culture
IGI Global, 2014

TALKS

"Takeaways from Major Software Supply Chain Attacks" (webinar)

Chainguard

May 12, 2026

[Webinar registration](#)

"Securing ML Pipelines: Way More Than You Wanted to Know" (workshop, with Kalle Westerling)

RSAC Conference 2026

March 2026 — San Francisco, CA

"Attacks Rewritten: Where Malware Enters the Build" (with Manfred Moser)

Chainguard Assemble 2026

March 17, 2026 — New York, NY

[Recording](#)

[Blog post](#)

"Dispelling the Myths of Advisory Feeds" (with Konstantinos "Gus" Evangelakos, Orca Security)

Chainguard Assemble 2026

March 17, 2026 — New York, NY

[Recording](#)

"Agent Omelets & Other Forbidden Techniques"

AI Native Dev DevCon Fall 2025

November 2025

[Recording](#)

"Frankenpipe: Bringing Secure ML Pipelines to Life"

All Things Open 2025

October 2025 — Raleigh, NC

[Recording](#)

[Session page](#)

"Deploy Secure Containers on Amazon EKS with Chainguard" (webinar, with Sai Vennam,

AWS)
Chainguard / AWS
August 2025
[Recording](#)

"Cheese Must Stand: Defending the Python Library Ecosystem"
PyCon US 2025
May 2025 — Pittsburgh, PA
[Recording](#)

"The Old, the New, and the Strange: Securing Deep Learning"
Chainguard Assemble 2025
March 2025 — San Francisco, CA
[Recording](#)

"Beyond Zero: Eliminating Vulnerabilities in the PyTorch Container Image" (lightning talk, with Dan Fernandez and Srishti Hegde)
PyTorch Conference 2024
September 2024 — San Francisco, CA
[Recording](#)
[Materials](#)

"Securing the Foundations of AI Applications"
JFrog SwampUP 2024
September 2024

"Zero CVE CUDA Chainguard Images"
Austin Kubernetes meetup
May 17, 2024
[Recording](#)

Notebooks for All: Accessibility & Jupyter Notebooks
JupyterCon 2023
Paris, France
[Description](#)

Day of Accessibility
Space Telescope Science Institute
Baltimore, MD
[Recordings](#)
[Event Page](#)

Accessible Notebooks Hackathon
Space Telescope Science Institute
2023, Online
[Event Page](#)
[Activity page](#)

[Event Recording](#)

"Haccessibility for All"

arXiv Accessibility Forum 2023

April 17, 2023

[View Recording](#)

STEM access: From author to reader

T_EX Hour Mini-Conference

2023, Online

[Event page](#)

"The Accessible Laboratory"

STEM Pedagogy Institute

2022, New York, NY

[View Curriculum](#)

"Web APIs with Python"

Week-long course, Digital Humanities Summer Institute

2018, 2019 - Victoria, British Columbia

[View Materials](#)

[#dhsiapi on Twitter](#)

"Rules of Thumb for Computer Security"

New York Amateur Computer Club

2017 — New York, NY

[Description](#)

[Presentation](#)

"Poster Session: DH Box"

Digital Humanities 2017

[Poster](#)

"Code and the Codex: E-books as Applications"

[E-book Revolution panel](#)

MLA Convention

2017 — Philadelphia, PA

[View slides](#)

[View notes](#)

"DH Box"

[Open Source Lit, Open Source Crit panel](#)

MLA Convention

2017 — Philadelphia, PA

"GC Digital Initiatives and the Digital Research Institute: Enhancing Student Experience and Access"

CUNY IT Conference
2016 — New York, NY

"Blind Hackers: Accessibility by Other Means"

a11y Accessibility Camp

2016 — New York, NY

[Watch video](#)

[View slides](#)

[View notes](#)

"DH Box: Building Cloud-Based Infrastructure for the Digital Humanities"

Canadian Society for Digital Humanities, Congress 2016

2016 — Calgary, Alberta

(Travel funding from National Endowment for the Humanities)

"Building the University Worth Fighting For: Tools for Pedagogical, Institutional, and Social Change"

HASTAC conference

2016 — Tempe, Arizona

(Travel award from GC Futures Initiative)

"Lightning Talk: DH Box"

NEH Project Director's Meeting

2015 — Washington, DC

"Introducing DH Box"

Media Res: Lightning Talks by NYC Graduate Students on DH Projects

2015 — New York, NY

"DH Box: A Push-Button Digital Humanities Laboratory"

METRO Conference

2015 — New York, NY

"DH Box: A Digital Humanities Laboratory in the Cloud"

CUNY DHI: Building a Digital Humanities Community

2015 — New York, New York

"Ebooks and the Digital Paratext: Emerging Trends in the Interpretation of Digital Media"

Digital Materials conference

2015 — Galway, Ireland

(Travel funding from GC Digital Initiatives)

"To Look upon Verdure": Transplantation and Cultivation in Mansfield Park"

Association for the Study of Literature and the Environment Conference

2015 — Moscow, ID

"Divining Disenchantment: Pneuma and Early Modern Science"

Warren Susman Graduate Conference

2015 — New Brunswick, NJ

"The Currency of Belief: Magical Realism and Fantasy in Neil Gaiman's American Gods"

International Conference for the Fantastic in the Arts

2015 — Orlando, FL

"DH Box: A Push-Button Digital Humanities Laboratory in the Cloud"

CUNY IT Conference

2014 — New York, New York

"The Present and Future of the Digital Graduate Center"

CUNY IT Conference

2014 — New York, New York

"Hangmen, Harlots, and Highwaymen: Narratives of Coercion and Resistance in the Eighteenth Century"

Northeast Modern Language Association Conference (NeMLA)

2013 — Boston, MA

CLIENT LIST

Astronomy Notebooks for All

[Space Telescope Science Institute](#)

2022 — 2024

Digital Humanities Research Infrastructure for Teaching (DHRIFT)

[The Graduate Center](#), City University of New York

2023 — 2024

Enthought Edge Platform Documentation

[Enthought](#)

2023

Google Summer of Documentation Initiative

[SigStore](#)

2023 — 2024

Badging infrastructure Development Project

[Hikma Studio](#)

2023 — 2024

AWARDS AND GRANTS

Carl Quinn Speaker Award

JFrog SwampUP 2024

[Award announcement](#)

Alumni and Doctoral Faculty Prize for the Most Distinguished Dissertation of the Year
Department of English, The Graduate Center, CUNY
2021

Donald and Mary Ellen Passantino Students with Disabilities Award
CUNY University Student Senate
2018

Milton F. Steinhardt Memorial Fellowship Award for Best Prospectus
Dissertation Year Prize, Graduate Center English Department
2018-2019

Public Humanities Fellowship
[Humanities New York](#)
Project: "Apps for Activists"
2018-2019

NYCDH Graduate Student Digital Project Award, "Futures Past Archive" ([Second place](#))
2016

Provost's Digital Innovation Grant, "Futures Past Archive"
2015-2016 and 2016-2017

Digital Writing Fellowship, Qwriting, Queens College
2016-2017

ARC Praxis Fellowship, Advanced Research Collaborative, CUNY Graduate Center
2014

Graduate Research Assistantship, David Greetham, CUNY Graduate Center
2012

Fulbright Fellowship, English Teaching Assistantship in Berlin, Germany
2009-2010

PRESS

[Notebooks For All: Making Jupyter Notebooks More Accessible](#)
Astrobites, 2023

[A Graduate Center Fellowship That Builds Digital Skills Also Yields Jobs for Alumni](#)
CUNY News, 2021

[Behind the Key — Patrick Smyth](#)
Phi Beta Kappa website, 2017

["Trump Budget Slashes NEH Funding: Why You Should Care"](#)
PC Magazine, 2017 (Article features the [NEH Impact Index](#).)

SERVICE

Student Representative, Interactive Technology and Pedagogy Student Liaison Committee
2016-2017 and 2017-2018

Student Representative, Graduate Center ADA Compliance Committee
2016-2017

Convener, GC Python User Group (PUG)
2015-2017

Member, English Student Association Website Committee
2015-2016

Chair, GC Digital Research Institute Curriculum Committee
2015-2017

Member, GC Digital Research Bootcamp Steering Committee
2015-2017

Convener, Utopian Studies Seminar, CUNY Graduate Center
2012-2015

PROJECTS

Project Lead, [NEH Impact Index](#)
2017
[Site](#) | [FAQ](#) | [Blog post](#) | [Twitter](#)
[PC Magazine Coverage](#)

Principal Investigator, Futures Past Archive
2015 — 2017

Maintainer, [Eloud screen reader](#)
2016 - 2017

Developer, [DH Box](#), CUNY Graduate Center
2014 — 2018

Platform Redesign, [Writing Studies Tree](#), CUNY Graduate Center
2014 — 2015

Contributing Historian, [More Than a Mapp](#), CUNY Graduate Center
2013 – 2014

Website Redesign, [Association of Independent Information Professionals](#) (AIIP)
2013-2014

Dialogue and Scripting, AI Patient Platform, MATRIC Research
2012 – 2013

CLASSES TAUGHT

Software Design Lab, MA Program in Digital Humanities, The Graduate Center, Spring 2019

Literature and Place: Imagining Utopia, Queens College, Spring 2014

Intro Composition (Information Technology theme), Queens College, Spring 2014

Intro Composition (Evolutionary Science theme), Queens College x2, Fall 2013

WORKSHOPS TAUGHT

"Securing the AI/ML Supply Chain" course launch (Chainguard, July–August 2024)

[Nonvisual Data Science Workshops](#), NumFOCUS small development grant (Iota School, January–March 2024, 400+ participants)

Getting Started with Python (Foundations for Research Computing, Columbia University, spring 2021)

"Pandas: The Bare Basics" (Columbia Python User Group, fall 2019)

"An Introduction to Static Sites with Jekyll" (Digital Humanities Research Institute, summer 2018)

"Twitterbots with the Twitter API" (Digital Humanities Research Institute, summer 2018)

"Python 201" (GC Digital Fellows, spring 2018)

"Open and Accessible: A Critical Distinction" (Teach @ CUNY Day, Summer 2017)

"Intro to Programming with Python" (GC Digital Research Institute, Fall 2016)

"Intro to HTML and CSS" (ITP Program, Fall 2016)

"Introduction to Web Frameworks with Flask" (GC Digital Research Institute, Summer 2016)

"Social Media Scraping with the Twitter API" (GC Digital Research Institute, Summer 2016)

"Introduction to the Command Line" (GC Digital Research Institute, Summer 2016)

"Collaboration and Writing Workflows with Git and GitHub" (GC Digital Fellows, Spring 2016)

"Social Media Scraping with the Twitter API" (NYC Digital Humanities Week, Spring 2016)

"Intro to Programming with Python" (NYC Digital Humanities Week, Spring 2016)

"Intro Python with DH Box" (ITP Program, Spring 2016)

"Responsive Design with Bootstrap" (ITP Program, Spring 2016)

"Intro to HTML and CSS" (ITP Program, Spring 2016)
"Automation and Scraping with the Twitter API" (GC Digital Research Bootcamp, Winter 2016)
"Intro to the Command Line" (GC Digital Fellows/CUNY GC Library, Fall 2015)
"Setting Up a Development Environment" (GC Digital Praxis Seminar, Fall 2015)
2x "Intro to Programming with Python" (CUNy Graduate Center Digital Fellows, Fall 2015)
"CSS and Bootstrap" (CUNY Graduate Center ITP Program, Fall 2015)
"Intro to Web Scraping with Python" (CUNY Graduate Center ITP Program, Spring 2015)
"Bootstrap Basics" (CUNY Graduate Center ITP Program, Spring 2015)
2x "Intro to Programming with Python," (CUNY Graduate Center Digital Fellows, Spring 2015)
"Getting Started with DH Box" (CUNY Graduate Center Digital Fellows, Fall 2014)
"Intro to Programming with Python," CUNY Graduate Center Digital Fellows, Fall 2014
2x "WordPress 2: Advanced Techniques," CUNY Graduate Center ITP Program, Fall 2014
"WordPress: Theming and Design," CUNY Graduate Center ITP Program, Spring 2013

TECHNICAL SKILLS

ML Infra: PyTorch, CUDA, NeMo, Pydantic AI, MCP
Agentic: Harness design, evals, multiagent flows
Ops: Docker, Unix, Git
Data Science: Jupyter, NLTK, NumPy/SciPy, spaCy, Pandas, scikit-learn
Web Frameworks: Flask, Django
CMS / Static Site Generators: Jekyll, Hugo, WordPress
Databases: MySQL, SQLite, PostgreSQL